

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, C++, PySpark

Tools/Frameworks: TensorFlow, PyTorch, FastAPI, Flask, Docker, Kubernetes, TensorRT, OpenAI API, AWS SageMaker, Git, Airflow, Hive, NiFi

Competence: Machine Learning, MLOps, Predictive Analytics, Time-Series Modeling, Supervised Learning, Feature Engineering, Experimental Design, Causal Inference, Data Engineering, Real-Time Systems, Data Visualization

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, enhancing model accuracy and training efficiency.
- Developed and implemented physics-informed hybrid loss functions to improve model stability and performance.
- Collaborated with PhD researchers to co-author a publication for the FLOMEKO conference.
- Built interactive dashboards for real-time data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Developed and deployed a vitiligo segmentation model with 94% accuracy, improving inference speed by 5x using TensorRT and FastAPI.
- Built a real-time tennis foul detection system, reducing compute costs and latency by 6x with optimized AI algorithms.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift and maintaining Kubernetes-based infrastructure.
- Designed and delivered educational sessions on machine learning and big data at universities, fostering knowledge sharing.
- Created POCs for clients using YOLO, few-shot learning, and CNNs to demonstrate AI capabilities.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website for AI-based services using Flask, HTML, and CSS, improving user engagement and functionality.

- Built an AI chatbot application leveraging OpenAI API and LLM frameworks to enhance user interaction and support.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data extraction and transformation.
- Automated data ingestion and cleaning workflows to streamline client analytics processes.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Developed a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Improved model accuracy by 4% through advanced feature engineering across MF, CC, and BP namespaces.

Razanalytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Built a real-time Bitcoin price forecasting system using time-series modeling and predictive analytics techniques.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Engineered the world's first AI-driven self-playing Chitralli Sitar using signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face

Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in machine learning, predictive analytics, and real-time decision systems.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence.
- Focused on data engineering, supervised learning, and experimental design.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python